

$$6.5) 2) B_0 = B = \{0, 1, 2, \dots\}, B_1 = \{-1, 0, 1, \dots\}$$

$$\text{so } B_n = \{-n, -(n-1), -(n-2), \dots, -1, 0, 1, 2, \dots\}$$

$$6.5) 3) 1) C = \bigcup_{n \in \mathbb{N}} B_n \subseteq A$$

$$\text{Supp } y \in \bigcup_{n \in \mathbb{N}} B_n \longrightarrow y \in A$$

$$\text{iff } \exists n \in \mathbb{N} (y \in B_n) \text{ iff } \exists f \in F (y \in f(B_n)) \text{ iff } \exists x \in B_n (f(x) = y)$$

$$\text{As } f: A \rightarrow A \text{ so } f(x) = y \in A.$$

$$2) \forall f \in F (\forall x \in C (f(x) \in C))$$

$$\text{Supp } f \in F \ \& \ x \in C. \longrightarrow f(x) \in C.$$

$$\text{Then } x \in \bigcup_{n \in \mathbb{N}} B_n \text{ so } \exists n \in \mathbb{N} (x \in B_n) \text{ so } f(x) \in f(B_n) \text{ so } f(x) \in \bigcup_{f \in F} f(B_n)$$

$$\text{so } f(x) \in B_{n+1} \text{ so } f(x) \in \bigcup_{n \in \mathbb{N}} B_n = C$$

$$3) \text{Supp } B \subseteq D \subseteq A \ \& \ D \text{ is closed under } F \quad C \subseteq D$$

$$\bigcup_{n \in \mathbb{N}} B_n \subseteq D$$

We will prove by induction that $\forall n \in \mathbb{N} (B_n \subseteq D)$, then $C \subseteq D$

Base case: $n=0$, then $B_0: B \subseteq D$ by Assumption.

$$\text{Induction step: Supp } \forall k \in \mathbb{N} (B_k \subseteq D) \longrightarrow B_n \subseteq D$$

$$\text{Supp } y \in B_n \longrightarrow y \in D$$

$$y \in \bigcup_{i \in \mathbb{N}} f(B_{n-1}) \iff \exists f \in F (y \in f(B_{n-1}))$$

$$\iff \exists x \in B_{n-1} (f(x) = y).$$

As by Inductive Hypothesis, $B_{n-1} \subseteq D$ so $x \in D$. Since D is closed, so $f(x) = y \in D$.

6.5) 6) The mistake is in the proof of C being closed. When $x \in C$ & $y \in C$ then $\exists n \in \mathbb{N} (x \in B_n)$ & $\exists m \in \mathbb{N} (y \in B_m)$ but it may be that $m \neq n$. So the set $\bigcup_{n \in \mathbb{N}} B_n$ is not necessarily closed under F . Example: Let $A = \mathbb{N}^+$, $B = \{1\}$, $F(a, b) = a + b$

$$\text{then } B_0 = B = \{1\}$$

$$B_1 = F(B_0 \times B_0) = \{2\}$$

$$B_2 = F(B_1 \times B_1) = \{4\}$$

$$\text{so } C = \{1, 2, 4, 8, \dots\}$$

$$\text{But } f(1, 2) = 1 + 2 = 3 \notin C.$$

A correct recursive Definition would be

$$\mathbf{6.5) 7)} \quad B_{n+1} = B_n \cup f(B_n \times B_n) \quad \text{--- (1)}$$

We can prove that using Definition (1), $C = \bigcup_{n \in \mathbb{N}} B_n$ is closed under f . It is helpful to first prove the following:

$$\forall m \leq n (x \in B_m \rightarrow x \in B_n) \quad \text{--- (2)}$$

Using Induction on m is tricky, so we will use Induction on n .

The base case is $n = 0$. It can also be $n = m$ which is sometimes called induction starting from m .

Base Case: $n = 0$ then $m = 0$ & $x \in B_0 \rightarrow x \in B_0$.

Induction Step: Supp $m \leq n$ be arbitrary & $x \in B_m \rightarrow x \in B_n$ --- (3)

We have to prove $x \in B_n \rightarrow x \in B_{n+1}$. Supp $x \in B_m$. By (3)

$x \in B_n$ so $x \in B_n \cup f(B_n \times B_n) = B_{n+1}$. Hence Proved.

Now we will prove that C is closed under f

Supp $x \in C, y \in C \longrightarrow f(x, y) \in C$

$\rightarrow \exists n \in \mathbb{N} (x \in B_n) \ \& \ \exists m \in \mathbb{N} (y \in B_m)$

Case 1) $n \geq m$. Then by ②, $y \in B_n$, so $x, y \in B_n \times B_n$ so

$f(x, y) \in f(B_n \times B_n)$ so $f(x, y) \in C$

Case 2) $m > n$. Then by ②, $x \in B_m$ & as above, similar reasoning shows $f(x, y) \in C$.

$\therefore$ The motivation for proving ② emerged as if $x \in B_n$ & $y \in B_m$, the 4 cases this leads to depend on whether $m \geq n$ or $m < n$.

If a statement is about multiple arbitrary variables, we can use induction on any one.

6.5) 11) let n be arbitrary. We will use induction on n

Supp $R^{n+1} = R^n \circ R \longrightarrow R^{m+(n+1)} = R^m \circ R^{n+1}$

$R^{m+(n+1)} = R^{m+n+1} = R^{m+n} \circ R$ ($\because$ By Recursive Def)

$= R^m \circ R^n \circ R$ ($\because$ By ①)

$= R^m \circ R^{n+1}$ ($\because$ By Recursive Def)

6.5) 12) (a) We use induction on n .

Base Case: $n=1$. $f' = f$ is function by Assumption.

Induction step: Supp $f^n: A \rightarrow A$ $\longrightarrow f^{n+1}: A \rightarrow A$

$f^{n+1} = f^n \circ f$ ($\because$ By Recursive Def)

f^n is function by ① & f is function by Assumption. Since composition of 2 functions is a function, so $f^{n+1}: A \rightarrow A$.

6.5) 12) (b) We use induction on n .

Base Case: $n=1$, then $f'(B) = B$, by definition

Induction step: (Not) Assume $f^n(B) = B_n$. Note that $f^{n+1}(B)$

$= f^n \circ f(B) = f^n(f(B)) = f^n(B_1)$ so we are unable to use

the Inductive Hypothesis. To use it, the key is using previous part (a).

$$\begin{aligned}
 f^{n+1}(a) &= f^{1+n}(a) = f \circ f^n(a) && \therefore \text{By part (a)} \\
 &= f(f^n(a)) \\
 &= f(b_n) && \therefore \text{By Inductive Hypothesis} \\
 &= b_{n+1}
 \end{aligned}$$

6.5) 13) (a) Here is how a loop forms when $\exists n \in \mathbb{Z}^+ (f^n(a) = a)$

$$\begin{aligned}
 \text{supp } n=3 &\rightarrow f(a) = b \\
 f^2(a) &= f(b) = c \\
 f^3(a) &= f(c) = a
 \end{aligned}$$

Now
 $f^4(a) = f(a) = b$ again so there will be only 3 elements in $\bigcup_{n \in \mathbb{N}} B_n$.

Also note from previous exercise $\forall n [f^n(a) = b_n]$. Also note that $\forall n \in \mathbb{N} [|B_n| = 1]$. This can be proved by Induction.

$$n=0 \rightarrow B_0 = \{a\} \text{ so } |B_0| = 1$$

$$n \geq 0 \rightarrow \text{supp } |B_n| = 1. \text{ Then } B_{n+1} = f(B_n). \text{ Since } |B_n| = 1, \frac{1}{f} [f(B_n)] = |B_{n+1}| > 1, \text{ then } f \text{ is not a function so } |B_{n+1}| = 1.$$

In this question, using Induction on n is difficult as there will be no way to relate to Inductive Hypothesis.

In other words, if we try to prove

$$f^1(a) = a \rightarrow C = \bigcup_{n \in \mathbb{N}} B_n \text{ is finite}$$

$$\wedge (f^n(a) = a \rightarrow C \text{ is finite}) \rightarrow (f^{n+1}(a) = a \rightarrow C \text{ is finite})$$

Then there is no way to relate $f^{n+1}(a) = a$ to Inductive Hypothesis as there is no element whose period is n . There is only a whose period is $n+1$. So Induction cannot be done on n .

There are 2 ways to prove this: With & Without Induction.

We know by example for $n=3$ that since elements go into a loop,

so the set $D = \{a, f(a), f^2(a), \dots, f^{n-1}(a)\}$ would be formed

by $\bigcup_{n \in \mathbb{N}} B_n$ when $f^n(a) = a$ again so a is periodic point.

Proof without Induction: Supp a is periodic point for f . So

$\exists n > 0$ ($f^n(a) = a$). We can show that C is finite using D .

Let D be as above. We will prove D is closed.

Supp $x \in D$ $\longrightarrow$ $f(x) \in D$

then $\exists 0 \leq k \leq n-1$ ($x = f^k(a)$)

Case 1) $0 \leq k < n-1$. Then $k+1 \leq n-1$ & $f(x) = f(f^k(a))$

$= f^{k+1}(a) \in D$

Case 2) $x = f^{n-1}(a)$, then $f(x) = f(f^{n-1}(a)) = f^n(a)$ but n is period so $f^n(a) = a \in D$.

So D is closed. Since C is closure so $C \subseteq D$. Since D is finite, so C is finite.

Proof By Induction is similar: We just let n be arbitrary.

Assume $f^n(a) = a$. We know $C = \bigcup_{n \in \mathbb{N}} B_n = B_0 \cup B_1 \cup B_2 \dots$

$= \{a\} \cup f(\{a\}) \cup f^2(\{a\}) = \{a, f(a), f^2(a), \dots\}$ so we

can show by Induction that $\forall m \in \mathbb{N}^+ [f^m(a) \in D]$ the base cases are

true for $1 \leq m \leq n-1$ & a . In other words, $a \in D$ & $\forall 1 \leq m \leq n-1$,

$f^m(a) \in D$ by Definition of D .

In Induction step, we can show that $\forall m \geq n-1$ ($f^m(a) \in D$)

Assume $f^m(a) \in D$, then $f^m(a) = f^k(a)$ for $1 \leq k \leq n-1$ or

$f^m(a) = a$ but m is not period as n is period so $k \leq n-1$.

Now we just use similar reasoning as in proof without Induction

to show that $f^{m+1}(a) = f^{k+1}(a) \in D$.

6.5) 13) (b) No, we might stop at example $f^n(a) = b$ where b has already occurred & $b \neq a$.

6.5) 15) Base Case: $n=1$. Then $R' \subseteq S'$ by assumption.

Induction step: $\text{supp } R^n \subseteq S^n$. — ①

$\text{supp } (x, y) \in R^{n+1}$ so $(x, y) \in R^n \circ R$ so $\exists z \in A ((x, z) \in R^n \wedge (z, y) \in R^n$

As $R \subseteq S$ & $R^n \subseteq S^n$ by ① so $(x, z) \in S^n$ & $(z, y) \in S^n$. so

$(x, y) \in S^n \circ S = S^{n+1}$.